

# Miaomiao Chen

Research Scientist @ AstraZeneca PLC

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## SUMMARY

Ph.D. in Chemistry with 7+ years of experience in machine learning modeling and large-scale data analysis, grounded in statistics and applied mathematics. Hands-on experience developing LLM-powered and agentic AI systems, with MLOps and EDA applied to multi-omics data for pharmaceutical and biomedical applications. **Fun fact:** China National Chess Champion (2011 2013 and 2014).

## EDUCATION

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|--------------------------------------------------------------------|----------------|
| Georgia Institute of Technology, MS in Computer Science (ML Track) | 2024 – Present |
| University of Virginia, PhD in Chemistry                           | 2024           |
| ShanghaiTech University, BS in Bioscience                          | 2019           |

## SKILLS

- **Programming:** Python, SQL, Spark
- **Machine Learning:** Transformers, LLMs, RL, Agentic AI (LangChain), PyTorch, RecSys (RAG, HSTU)
- **MLOps & Deployment:** EDA, Domino (Kubernetes), FastAPI services, Bazel

## EXPERIENCE

**Research Scientist, Clinical Proteomics, AstraZeneca Pharmaceuticals PLC, MD** Jul 2024 – Present

- Led development of domain-specialized AI agents leveraging LangChain, RAG, and MonetDB to automate proteomics analytics, delivering a 3× acceleration in pattern discovery across AstraZeneca’s clinical cancer drug portfolio.
- Designed and productionized patient embedding models using NMF and two-tower architectures within a full MLOps pipeline, driving a 30% ROC-AUC improvement for clinical drug-response prediction.
- Fine-tuned encoder-decoder foundation models via transfer learning on large-scale proteomics and multi-omics clinical trial data, achieving a 20% ROC-AUC gain in response-rate prediction.
- Deployed and operated production-grade Python/Flask services on Domino (Kubernetes) to standardize clinical proteomics QC, integrating statistical analysis and visualization, resulting in a 20% efficiency uplift and adoption as a core team workflow.

**PhD Researcher, University of Virginia, VA** 2019 – 2024

- Designed ML models (RF, XGBoost, LSTM) for drug candidate down-selection, achieving 85.3% accuracy, shortening experimental timelines by months, and identifying 2 of 10 effective candidates, with results published in *JACS*.
- Performed large-scale EDA and designed A/B testing for anti-inflammation drug studies, contributing to publications including first-authored work in *PNAS*.

## SELECTED PROJECTS

**Krono: Voice-Activated AI News Agent** 🗣️ 🎧 2025 – Present

- Designed a LangGraph-based multi-agent system for intent detection, tool routing, retrieval, and generation, supporting real-time, fully voice-driven news exploration with persistent conversational context and natural follow-ups.
- Implemented a lightweight, thread-based audio interaction architecture enabling real-time interruption and refinement (<100 ms response), achieving high concurrency and ~60% lower memory usage than multiprocessing approaches.
- Post-trained and deployed Qwen-1.5B using GRPO with vLLM as an internal LLM for large-scale news classification, reducing external LLM API usage by 60% and inference costs by 50%.
- Launched a production voice-based AI news agent that performs real-time retrieval, ranking, and summarization of market news, reducing user lookup time by ~70%, while integrating SenseVoice ASR and AI agentic pipelines.

**Reproduce GPT-2 with Fine Tune** 🗣️ 📝 2024 – Present

- Built and trained a custom GPT-2 style model (12 transformer layers, 384-dim embeddings), achieving a 70% reduction in validation loss and convergence close to the OpenAI GPT-2 (124M) benchmark.
- Achieved strong generalization on the HellaSwag commonsense reasoning benchmark (26% accuracy), demonstrating robust and stable performance over 3,800+ training steps.
- Leveraged QLoRA for efficient lightweight fine-tuning, improving multilingual response quality (train loss 0.77) using 1,000 samples from the Guanaco dataset across 5 languages.

**Reinforcement Learning of Flappy Bird Game** 🎮 2024 – 2025

- Led the design of a DQN-based reinforcement learning pipeline for training an autonomous Flappy Bird agent, incorporating experience replay and  $\epsilon$ -greedy exploration to stabilize learning.
- Implemented Double and Dueling DQN architectures, accelerating reward convergence by 80% and significantly improving training stability over vanilla DQN.
- Built a configurable, hyperparameter-driven RL framework (YAML-based) adaptable across environments, enabling the agent to pass 300+ obstacles and sustain ~16.5% average reward improvement between learning milestones.